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**THE PRINCIPLE OF STATIONARY PHASE
FOR THE FOURIER TRANSFORM OF D -MODULES**

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THE PRINCIPLE OF STATIONARY PHASE FOR THE FOURIER TRANSFORM OF D -MODULES

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We show that the formal germ at the infinity of the Fourier transform of a holonomic D -module depends only on the formal germ of the D -module at its singular points and at the infinity.

1. Introduction

The stationary phase approximation is a basic principle of asymptotic analysis, exemplified by the oscillatory integral

$$I(t') = \int g(t) e^{it' f(t)} dt.$$

If the derivative of $f(t)$ does not vanish at any point in $\text{Supp}(f)$, then $I(t')$ is rapidly decreasing at ∞ . If $f(t)$ has only finitely many critical points in $\text{Supp}(f)$, the major contribution to the value of the integral $I(t')$ for large t' comes from neighborhoods of those critical points. More generally, consider the integral

$$I(t') = \int_{a(t')}^{b(t')} g(t, t') e^{if(t, t')} dt,$$

where all the functions are real-valued. Under certain conditions, for $t' \rightarrow \infty$,

$$I(t') = \sum_{f_i(t, t')=0} \left(g(t, t') \sqrt{\frac{2\pi}{|f_{tt}(t, t')|}} e^{if(t, t') + \frac{i\pi}{4} \text{sgn} f_{tt}(t, t')} + o\left(\frac{g(t, t')}{\sqrt{|f_{tt}(t, t')|}}\right) \right).$$

The classical principle of stationary phase outlined above relates to the real Fourier transform. To study Deligne's ℓ -adic Fourier transform, Gérard Laumon [1987] introduced a corresponding principle of stationary phase and the local ℓ -adic Fourier transform. (See [Katz 1988] for a good exposition.) We are interested in the D -module case.

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We fix a field k of characteristic 0 and use the following notations:

- (1) Let p_1, p_2 be the projections $\text{Spec } k[t, t'] = \mathbb{A}_k^1 \times_k \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^1$, and let $\bar{p}_1, \bar{p}_2$ be the projections $\mathbb{P}_k^1 \times_k \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$. Let $\alpha : \mathbb{A}_k^1 \hookrightarrow \mathbb{P}_k^1$ and $\mu : \mathbb{A}_k^1 \times_k \mathbb{A}_k^1 \hookrightarrow \mathbb{P}_k^1 \times_k \mathbb{P}_k^1$ be the inclusions.
- (2) For any $x \in k$, let $t_x = t - x$ and $t'_x = t' - x$. Let $t_\infty = 1/t = z$, $t'_\infty = 1/t' = z'$ and $\eta' = \text{Spec } k(t')$. For any $x \in k \cup \{\infty\}$, let $\eta_x = \text{Spec } k((t_x))$, $\eta'_x = \text{Spec } k((t'_x))$.
- (3) For any $x, y \in k \cup \{\infty\}$, let $k((t_x, t'_y))$ be the field of the formal Laurent series $\sum_{i, j \gg -\infty} a_{ij} t_x^i t_y^j$, $a_{ij} \in k$. For any $k((t_x))$ -vector space M , let

$$M((t'_y)) = M \otimes_{k((t_x))} k((t_x, t'_y)).$$
- (4) Denote by $\mathcal{L}$ the rank-one connection $(\mathbb{O}_{\mathbb{A}_k^1}, d+dt)$ on $\mathbb{A}_k^1$. Then $\mathcal{L}$ corresponds to the D -module $\mathbb{O}_{\mathbb{A}_k^1} \cdot e^t$ on $\mathbb{A}_k^1$. So $\mathcal{L}$ is a substitute of e^{it} in classical Fourier analysis. Let X be a scheme. Any section $f \in \mathbb{O}(X)$ defines a morphism $\phi : X \rightarrow \mathbb{A}_k^1$ and let $\mathcal{L}_f = \phi^* \mathcal{L}$.

Let $\mathcal{M}$ be a vector bundle with a connection ∇ on a nonempty open subscheme U of $\mathbb{A}_k^1$ and let $i : U \hookrightarrow \mathbb{A}_k^1$ and $j : U \rightarrow \mathbb{P}_k^1$ be the inclusions. The connection ∇ on $\mathcal{M}$ can be extended to a connection $i_* \nabla$ on $i_* \mathcal{M}$ and a connection $j_* \nabla$ on $j_* \mathcal{M}$. The global (geometric) Fourier transform of the D -module $i_* \mathcal{M}$ on $\mathbb{A}_k^1$ is defined to be

$$\mathcal{F}(i_* \mathcal{M}) = p_{2+}(p_1^* i_* \mathcal{M} \otimes_{\mathbb{O}_{\mathbb{A}_k^1} \times \mathbb{A}_k^1} \mathcal{L}_{t't'})[1],$$

where $\otimes$ and p_{2+} are derived functors of D -modules. This definition is analogous to

$$\widehat{f}(t') = \int f(t) e^{itt'} dt.$$

More precisely, we have

$$\begin{aligned} \mathcal{F}(i_* \mathcal{M}) &\cong R^1 p_{2*}(p_1^* i_* \mathcal{M} \xrightarrow{p_1^* i_* \nabla + t' dt} p_1^*(\Omega_{\mathbb{A}_k^1}^1 \otimes_{\mathbb{O}_{\mathbb{A}_k^1}} i_* \mathcal{M})) \\ &\cong \alpha^* \alpha_* R^1 p_{2*}(p_1^* i_* \mathcal{M} \xrightarrow{p_1^* i_* \nabla + t' dt} p_1^*(\Omega_{\mathbb{A}_k^1}^1 \otimes_{\mathbb{O}_{\mathbb{A}_k^1}} i_* \mathcal{M})) \\ &\cong \alpha^* R^1 \bar{p}_{2*} \mu_* (p_1^* i_* \mathcal{M} \xrightarrow{p_1^* i_* \nabla + t' dt} p_1^*(\Omega_{\mathbb{A}_k^1}^1 \otimes_{\mathbb{O}_{\mathbb{A}_k^1}} i_* \mathcal{M})) \\ &\cong \alpha^* R^1 \bar{p}_{2*} (\bar{p}_1^* j_* \mathcal{M} \otimes \mu_* \mathbb{O}_{\mathbb{A}_k^1 \times \mathbb{A}_k^1} \xrightarrow{\bar{p}_1^* j_* \nabla + t' dt} \bar{p}_1^*(\Omega_{\mathbb{P}_k^1}^1 \otimes j_* \mathcal{M}) \otimes \mu_* \mathbb{O}_{\mathbb{A}_k^1 \times \mathbb{A}_k^1}). \end{aligned}$$

Consider the complex

$$(*) \quad (\bar{p}_1^* j_* \mathcal{M} \otimes \mu_* \mathbb{O}_{\mathbb{A}_k^1 \times \mathbb{A}_k^1} \xrightarrow{\bar{p}_1^* j_* \nabla + t' dt} \bar{p}_1^*(\Omega_{\mathbb{P}_k^1}^1 \otimes j_* \mathcal{M}) \otimes \mu_* \mathbb{O}_{\mathbb{A}_k^1 \times \mathbb{A}_k^1}).$$

We have

$$\mathcal{F}(i_* \mathcal{M})|_{\eta_{\infty'}} = R^1 \bar{p}_{2*} (*)|_{\eta_{\infty'}}.$$

To study $\mathcal{F}(i_*\mathcal{M})|_{\eta_{\infty'}}$, one needs to study $R^1\bar{p}_{2*}(*)|_{\mathrm{Spf} k[[z']]}.$ The complex $(*)$ involves quasicoherent sheaves that may not be coherent sheaves. To study the localization of $(*)$ on $\mathrm{Spf} k[[z']]$, we need to transform them into coherent sheaves. For this reason, Bloch and Esnault [2004] rewrote $(*)$ in terms of the cohomology of a complex of coherent modules. They found a good lattice pair $\mathcal{V}, \mathcal{W}$ of the connection $j_*\mathcal{M}$ such that $(\bar{p}_1^*j_*\nabla + t'dt)(\bar{p}_1^{*\mathcal{V}}) \subset \bar{p}_1^*(\Omega_{\mathbb{P}_k^1}^1(T) \otimes \mathcal{W})$ and the inclusion of complexes

$$(\bar{p}_1^{*\mathcal{V}} \xrightarrow{\bar{p}_1^*j_*\nabla + t'dt} \bar{p}_1^*(\Omega_{\mathbb{P}_k^1}^1(T) \otimes \mathcal{W})) \subset (*)$$

is a quasi-isomorphism. Here $T = \mathbb{P}_k^1 - U$. However, for any good lattice pair $\mathcal{V}, \mathcal{W}$ of the connection $j_*\mathcal{M}$, the conditions above do not hold, because the differential form $t'dt$ is singular on $\mathbb{P}_k^1 \times \{\infty\} \cup \{\infty\} \times \mathbb{P}_k^1$. We only have

$$(\bar{p}_1^*j_*\nabla + t'dt)(\bar{p}_1^{*\mathcal{V}}) \subset \bar{p}_1^*(\Omega_{\mathbb{P}_k^1}^1(T) \otimes (\mathcal{W} + \mathcal{V}(\{\infty\}))) (\mathbb{P}_k^1 \times \{\infty\})$$

and a subcomplex

$$(1-1) \quad (\bar{p}_1^{*\mathcal{V}} \xrightarrow{\bar{p}_1^*j_*\nabla + t'dt} \bar{p}_1^*(\Omega_{\mathbb{P}_k^1}^1(T) \otimes (\mathcal{W} + \mathcal{V}(\{\infty\}))) (\mathbb{P}_k^1 \times \{\infty\}))$$

of $(*)$. This inclusion of complexes $(1-1) \subset (*)$ is still not a quasi-isomorphism. Using Deligne's construction of good lattice pairs, we find a good lattice pair $\mathcal{V}, \mathcal{W}$ of $j_*\mathcal{M}$ in Lemma 2.3 such that $(1-1)|_{\mathbb{P}_k^1 \otimes_k k(t')} \subset (*)|_{\mathbb{P}_k^1 \otimes_k k(t')}$ is a quasi-isomorphism. From this, we get the following stationary phase formula.

Theorem 1.1. *Let $\mathcal{M}$ be a vector bundle with a connection ∇ on a nonempty open subscheme U of $\mathbb{A}_k^1$, and let $i : U \hookrightarrow \mathbb{A}_k^1$ be the inclusion. Suppose all points in $\mathbb{A}_k^1 - U$ are k -rational. Then the natural map*

$$(1-2) \quad \mathcal{F}(i_*\mathcal{M})|_{\eta_{\infty}} \rightarrow \bigoplus_{x \in \mathbb{A}_k^1 - U} \mathrm{coker}((\mathcal{M}|_{\eta_x})((z')) \xrightarrow{z'\partial_{t_x} + 1} (\mathcal{M}|_{\eta_x})((z'))) \\ \oplus \mathrm{coker}((\mathcal{M}|_{\eta_{\infty}})((z')) \xrightarrow{z'\partial_z - \frac{1}{z^2}} (\mathcal{M}|_{\eta_{\infty}})((z')))$$

is an isomorphism of formal connections on $k((z'))$.

The direct summands on the right side of (1-2) induce the definition of local Fourier transforms for formal connections.

The paper is organized as follows. In Section 2, we discuss the good lattice pairs of connections on a smooth curve. Passing to the stalks, we discuss the good lattice pairs of connections on a discrete valuation field. In Section 3, we prove the stationary phase formula using proper base change theorem between formal schemes.

2. Good lattice pairs

Let X be a smooth algebraic curve over k and $j : X \hookrightarrow \bar{X}$ the smooth compactification. Let $\mathcal{F}$ be a vector bundle on X with a connection ∇ . Set $\Sigma = \bar{X} - X$. A pair of good lattices $\mathcal{V}, \mathcal{W}$ of $j_*\mathcal{F}$ is a pair of vector bundles on $\bar{X}$ which extends $\mathcal{F}$ and satisfies the following conditions:

- (1) $\mathcal{V} \subset \mathcal{W} \subset j_*\mathcal{F}$.
- (2) $\nabla(\mathcal{V}) \subset \Omega_{\bar{X}}^1(\Sigma) \otimes \mathcal{W}$.
- (3) For any effective divisor D supported on Σ , the inclusion of complexes

$$(\mathcal{V} \xrightarrow{\nabla} \Omega_{\bar{X}}^1(\Sigma) \otimes \mathcal{W}) \rightarrow (\mathcal{V}(D) \xrightarrow{\nabla} \Omega_{\bar{X}}^1(\Sigma) \otimes \mathcal{W}(D))$$

is a quasi-isomorphism. Taking the direct limit with respect to D , we get a quasi-isomorphism:

$$(\mathcal{V} \xrightarrow{\nabla} \Omega_{\bar{X}}^1(\Sigma) \otimes \mathcal{W}) \rightarrow (j_*\mathcal{F} \xrightarrow{\nabla} \Omega_{\bar{X}}^1 \otimes j_*\mathcal{F}).$$

The existence of good lattice pairs can be passed to the stalks. So we only need to consider the local case: good lattice pairs of connections on a discrete valuation field.

Let K be a discrete valuation field with the valuation v . Let A be the valuation ring, t a uniformizer, and ∂ a continuous derivation on K such that $\partial(t) = 1$ and $\partial(A) \subseteq A$.

Definition 2.1. A connection on K (of rank k , where k is finite) is a k -dimensional vector space M over K with an additive map $\partial : M \rightarrow M$ satisfying $\partial(fm) = f\partial(m) + \partial(f)m$ for any $f \in K$ and $m \in M$.

Let r be the rank of the connection M . Set $\tau = t\partial$. There exists a cyclic element $v \in M$, in the sense that the elements $\tau^i v$, for $0 \leq i \leq r-1$, form a basis of M over K . We have

$$\tau^r v = \sum_{0 \leq i \leq r-1} a_i \tau^i v$$

for some $a_i \in K$. The Newton polygon $N(M)$ of M is the convex hull of

$$\{(u, v) \mid u \leq i, v \geq v(a_i)\}$$

in the plane $\mathbb{R}^2$. The slopes of M are the slopes of nonvertical edges of $N(M)$, and we eliminate the slope 0 if the horizontal edge is situated in $u \leq 0$. The slopes are independent of the choice of the cyclic elements. The sum of all the slopes of M is called the irregularity of M , and is denoted by $i(M)$. Then

$$i(M) = \max_{0 \leq i \leq r} (0, -v(a_i)).$$

A *lattice* of M is a finitely generated A -submodule V of M that spans M . For any artinian A -module V , the length of V is denoted by $\ell(V)$.

Definition 2.2. A pair of lattices V, W of (M, ∂) is called *good* if the following conditions are satisfied

- (1) $V \subset W \subset M$.
- (2) $\partial V \subset (1/t)W$.
- (3) For any $i \in \mathbb{N}$, the natural inclusion of complexes

$$\left(V \xrightarrow{\partial} \frac{1}{t} W \right) \rightarrow \left(\frac{1}{t^i} V \xrightarrow{\partial} \frac{1}{t^{i+1}} W \right)$$

is a quasi-isomorphism.

Note that if V, W is a good lattice pair, so is $(1/t^i)W, (1/t^i)V$ for any $i \in \mathbb{N}$.

Condition (3) above is equivalent to the following:

- (3') For any $i \in \mathbb{N}$, the map

$$\frac{1}{t^i} V \Big/ \frac{1}{t^{i-1}} V \xrightarrow{\text{gr}_i \partial} \frac{1}{t^{i+1}} W \Big/ \frac{1}{t^i} W$$

induced by ∂ is an isomorphism.

One can show that $i(M) = \ell(W/V)$.

Lemma 2.3. Let $k \hookrightarrow k'$ be an extension of fields of characteristic 0. Let ∂_t be the natural derivation on $k(t)$ and on $k'(t)$. The variable t defines a discrete valuation v on $k(t)$ and $k'(t)$. Let A and A' be their discrete valuation rings, respectively. Suppose c is an element in k' which is not algebraic over k . Let M be a connection on $k(t)$, and let M_c be the connection on $k'(t)$ whose underlying space is the $k'(t)$ -vector space $M \otimes_{k(t)} k'(t)$, and with the operation ∂_t defined by

$$\partial_t(m \otimes f) = \partial_t(m) \otimes f + m \otimes \partial_t(f) - m \otimes \frac{c}{t^2}$$

for any $m \in M$ and any $f \in k'(t)$. Then there exists a good lattice pair $\mathcal{V}, \mathcal{W}$ of M , such that $\mathcal{V} \otimes_A A', (\mathcal{W} + (1/t)\mathcal{V}) \otimes_A A'$ is also a good lattice pair of the connection M_c on $k'(t)$.

Proof. Set $r = \text{rk} M$. Choose a cyclic element v of M . Let ε be the basis $\{\tau^i v \mid 0 \leq i \leq r-1\}$ of M over $k(t')$. We have $\tau^r v = \sum_{0 \leq i < r} a_i \tau^i v$ for some $a_i \in K$. The irregularity $i(M)$ of M is $\max_{0 \leq i < r} (0, -v(a_i))$. Consider the Newton polygon of the differential operator $\tau^r - \sum_{0 \leq i \leq r-1} a_i \tau^i$. Let j be the integer such that $(j, v(a_j))$ is a vertex of this Newton polygon, and such that the slopes of

this Newton polygon on the right side (respectively left side) of $(j, v(a_j))$ is > 1 (respectively ≤ 1). Set $a_r = 1$. Then we have

$$\begin{aligned} v(a_{j+i}) - v(a_j) &> i \text{ for any } 1 \leq i \leq r - j, \\ v(a_{j-i}) - v(a_j) &\geq -i \text{ for any } 0 \leq i \leq j. \end{aligned}$$

Then

$$(2-1) \quad v(a_j) - j = \min_{0 \leq i \leq r} (v(a_i) - i).$$

The matrix of the differential operator τ with respect to the basis ε is

$$\Gamma = \begin{pmatrix} 0 & & a_0 \\ 1 & & a_1 \\ & \ddots & \vdots \\ & & 1 & a_{r-1} \end{pmatrix}.$$

The characteristic polynomial of Γ is $\lambda^r - \sum_{0 \leq i \leq r-1} a_i \lambda^i$. Let

$$\Lambda = \text{diag}\{1, \dots, 1, t, \dots, t^{r-j+i(M)+v(a_j)}\},$$

and let $e = \varepsilon \Lambda = \{e_i \mid 0 \leq i < r\}$. Set $l = j - v(a_j) - i(M) \geq 0$. Then the matrix of the differential operator τ with respect to the basis e is

$$\Gamma' = \begin{pmatrix} 0 & & & & t^{r-l}a_0 \\ 1 & & & & t^{r-l}a_1 \\ & \ddots & & & \vdots \\ & & 1 & & t^{r-l}a_{l-1} \\ & & & \frac{1}{t} & t^{r-l-1}a_l \\ & & & & \ddots \\ & & & & \frac{1}{t} & a_{r-1} \end{pmatrix} + \text{diag}\{0, \dots, 0, 1, \dots, r-l\}.$$

Let $P(\lambda) = \lambda^r - \sum_{0 \leq i \leq r-1} a'_i \lambda^i$ be the characteristic polynomial of Γ' . Since

$$\Gamma' = \Lambda^{-1} \Gamma \Lambda + \text{diag}\{0, \dots, 0, 1, \dots, r-l\},$$

we have

$$a'_i - a_i \in \sum_{i < j < r} \mathbb{Z} a_j + \mathbb{Z} \subset K.$$

So

$$\max\{0, -v(a'_i) \mid 0 \leq i < r\} = \max\{0, -v(a_i) \mid 0 \leq i < r\} = i(M).$$

Write $P(\lambda) = t^{-i(M)} \sum_i b_i \lambda^i$, $b_i \in K$. Then $b_i \in A$ and $v(b_i) = 0$ for at least one i . The residue polynomial $\sum_i \bar{b}_i \lambda^i$ of $\sum_i b_i \lambda^i$ is nonzero. For almost all $n \in \mathbb{Z}$,

$\sum_i \bar{b}_i (-n)^i \neq 0$. In this case, we have

$$-v(\det(n + \Gamma')) = -v((-1)^r P(-n)) = -v\left(t^{-i(M)} \left(\sum_i b_i (-n)^i\right)\right) = i(M).$$

Then, for almost all $n \in \mathbb{Z}$,

$$(2-2) \quad i(M) = -v(\det(n + \Gamma')).$$

Let V be the lattice of M generated by e . Define

$$(2-3) \quad [(n + \Gamma')V : V] = \ell((n + \Gamma')V + V/V) - \ell((n + \Gamma')V + V/(n + \Gamma')V).$$

By [Deligne 1970, p. 48, Proposition 2], we have

$$(2-4) \quad [(n + \Gamma')V : V] = -v(\det(n + \Gamma')).$$

Let W be the lattice of M generated by

$$e_0, \dots, e_{l-1}, \frac{1}{t}e_l, \dots, \frac{1}{t}e_{r-1}.$$

Then $\ell(W/V) = r - l$. Since $((n + \Gamma')V + V)/W$ is an artinian A -module generated by the single element

$$x = \sum_{0 \leq i \leq l-1} a_i t^{r-l} e_i + \sum_{l \leq i \leq r-1} a_i t^{r-1-i} e_i = \sum_{0 \leq i \leq l-1} a_i t^{r-l} e_i + \sum_{l \leq i \leq r-1} a_i t^{r-i} \frac{1}{t} e_i.$$

For any i , we have $i(M) \geq -v(a_i)$ and $v(a_j) - j \leq v(a_i) - i$. Then

$$v(t^{i(M)+l-r} a_i t^{r-l}) \geq 0 \quad \text{and} \quad v(t^{i(M)+l-r} a_i t^{r-i}) \geq v(t^{i(M)+l-r} a_j t^{r-j}) = 0.$$

Then the annihilator of x in $((n + \Gamma')V + V)/W$ is $t^{i(M)+l-r}$. So

$$\ell((n + \Gamma')V + V/W) = i(M) + l - r.$$

Then

$$(2-5) \quad \ell((n + \Gamma')V + V/V) = \ell(W/V) + \ell((n + \Gamma')V + V/W) = i(M).$$

Comparing this equality with (2-2), (2-3), and (2-4), we get

$$\ell((n + \Gamma')V + V/(n + \Gamma')V) = 0$$

for almost $n \in \mathbb{Z}$, that is, $(n + \Gamma')V \supset V$ for almost all $n \in \mathbb{Z}$.

The A -module

$$(n + \Gamma')V + \frac{1}{t}V \Big/ \frac{1}{t}V$$

is artinian and is generated by one element x whose annihilator is

$$t^{i(M)+l-r} = t^{j-v(a_j)-r}.$$

Then

$$(2-6) \quad \ell\left((n + \Gamma')V + \frac{1}{t}V \Big/ V\right) = \ell\left((n + \Gamma')V + \frac{1}{t}V \Big/ \frac{1}{t}V\right) + \ell\frac{1}{t}V \Big/ V \\ = j - v(a_j) = \sum_{\lambda: \text{slope of } M} \max(\lambda, 1).$$

The matrix of the differential operator τ with respect to the basis ε of M_c is $\Gamma - c/t$. The characteristic polynomial $P'(\lambda)$ of $\Gamma - c/t$ is

$$P'(\lambda) = \left(\lambda + \frac{c}{t}\right)^r - \sum_{0 \leq i < r} a_i \left(\lambda + \frac{c}{t}\right)^i.$$

Write $P'(\lambda) = \lambda^r + \sum_{0 \leq i < r} b_i \lambda^i$ for some $b_i \in k'(t)$. Then

$$b_0 = \left(\frac{c}{t}\right)^r - \sum_{0 \leq i < r} a_i \left(\frac{c}{t}\right)^i = \frac{a_j}{t^j} \left(\frac{1}{a_j t^{r-j}} c^r - \sum_{0 \leq i < r} \frac{a_i}{a_j t^{i-j}} c^i \right).$$

By (2-1), we have

$$\frac{1}{a_j t^{r-j}} c^r - \sum_{0 \leq i < r} \frac{a_i}{a_j t^{i-j}} c^i \in A[c],$$

and its residue in k' is a nonzero polynomial over k of c . Since c is not algebraic over k , this residue is nonzero. Then we have

$$v(b_0) = v\left(\frac{a_j}{t^j}\right) = v(a_j) - j.$$

Also by (2-1), we have $v(b_i) \geq v(b_0)$ for any $0 \leq i < r$. So

$$\max_{0 \leq i < r} (0, -v(b_i)) = j - v(a_j) = i(M_c).$$

The matrix of the differential operator τ with respect to the basis e of M_c is $\Gamma'' = \Gamma' - c/t$. Write the characteristic polynomial of Γ'' as $\lambda^r + \sum_{0 \leq i < r} b'_i \lambda^i$ for some $b'_i \in k'(t)$. By a similar proof as above, we have

$$\max_{0 \leq i < r} (0, -v(b'_i)) = \max_{0 \leq i < r} (0, -v(b_i)) = i(M_c).$$

For almost $n \in \mathbb{Z}$, we have

$$-v(\det(n + \Gamma'')) = i(M_c).$$

Let $V' = V \otimes_A A'$. We have $(n + \Gamma'')V' + V' \subseteq \frac{1}{t}V' + \Gamma'V'$; therefore So

$$(2-7) \quad \ell((n + \Gamma'')V' + V' / V') \leq \ell\left(\frac{1}{t}V' + \Gamma'V' \Big/ V'\right).$$

Since $A \rightarrow A'$ is flat and $k \otimes_A A' = k'$, for any artinian A -module M , one can prove $\ell(M) = \ell(M \otimes_A A')$. Since $(1/t)V + \Gamma'V/V$ is an artinian A -module, by (2-6),

we have

$$(2-8) \quad \ell\left(\frac{1}{t}V' + \Gamma'V' / V'\right) = \ell\left(\frac{1}{t}V + \Gamma'V / V\right) = j - v(a_j).$$

By (2-4), we have, for almost $n \in \mathbb{Z}$,

$$\begin{aligned} \ell((n + \Gamma'')V' + V' / V') &\geq \ell((n + \Gamma'')V' + V' / V') - \ell((n + \Gamma'')V' + V' / (n + \Gamma'')V') \\ &= -v(\det(n + \Gamma'')) = j - v(a_j). \end{aligned}$$

Comparing this inequality with (2-7) and (2-8), we have for almost $n \in \mathbb{Z}$,

$$\begin{aligned} \ell((n + \Gamma'')V' + V' / V') &= j - v(a_j); \\ \ell((n + \Gamma'')V' + V' / (n + \Gamma'')V') &= 0; \\ (2-9) \quad (n + \Gamma'')V' + V' &= \frac{1}{t}V' + \Gamma'V' = \left(\frac{1}{t}V + \Gamma'V\right) \otimes_A A'. \end{aligned}$$

So for almost $n \in \mathbb{Z}$, $(n + \Gamma'')V' \supseteq V'$. Let $e' = (1/t^N)e$. The matrix of τ with respect to the basis e' of M (respectively M'_c) is $\Gamma_1 := \Gamma' - N$ (respectively $\Gamma_2 := \Gamma'' - N$). Let $\mathcal{V} = (1/t^N)V$ and let $\mathcal{V}' = (1/t^N)V'$. Choose N large enough so that for any $n \leq 0$, we have

$$(n + \Gamma_1)\mathcal{V} \supset \mathcal{V} \quad \text{and} \quad (n + \Gamma_2)\mathcal{V}' \supseteq \mathcal{V}'.$$

Let $\mathcal{W} = \Gamma_1\mathcal{V}$. By (2-9), we have $\Gamma_2\mathcal{V}' = (\mathcal{W} + (1/t)\mathcal{V}) \otimes_A A'$. Let's prove $\mathcal{V}, \mathcal{W}$ is a good lattice of M now. We only need to verify condition (3') for any $i \in \mathbb{N}$. Conjugating by $1/t^i$, the A -linear map

$$\text{gr}_i \tau : \frac{1}{t^i}\mathcal{V} / \frac{1}{t^{i-1}}\mathcal{V} \rightarrow \frac{1}{t^i}\mathcal{W} / \frac{1}{t^{i-1}}\mathcal{W}$$

can be identified with the A -linear map

$$\text{gr}_0 \tau - i = \Gamma_1 - i : \mathcal{V}/t\mathcal{V} \rightarrow \mathcal{W}/t\mathcal{W}.$$

Since $(\Gamma_1 - i)\mathcal{V} \supset \mathcal{V}$, we have

$$(\Gamma_1 - i)\mathcal{V} = (\Gamma_1 - i)\mathcal{V} + \mathcal{V} \supset \Gamma_1\mathcal{V} = \mathcal{W}.$$

So $\Gamma_1 - i : \mathcal{V}/t\mathcal{V} \rightarrow \mathcal{W}/t\mathcal{W}$ is surjective. But the domain and the range of $\text{gr}_i \tau$ are artinian A -modules of the same length r , so $\text{gr}_0 \tau - i$ is an isomorphism and so is $\text{gr}_i \tau$. This proves $\mathcal{V}, \mathcal{W}$ is a good lattice pair of M . Repeating the proof, we conclude that $\mathcal{V} \otimes_A A', (\mathcal{W} + (1/t)\mathcal{V}) \otimes_A A'$ is a good lattice pair of M'_c . $\square$

Remark 2.4. Lemma 2.3 is the main technical lemma for the proof of the stationary phase principle in the next section. Lemma 2.3 also allows us to choose a good lattice pair $\mathcal{V}, \mathcal{W}$ of M such that

$$(2-10) \quad \dim_k \left(\mathcal{W} + \frac{1}{t}\mathcal{V} / \mathcal{V} \right) = \sum_{\lambda: \text{slope of } M} \max(\lambda, 1).$$

Formula (2-10) is easily seen to give a new proof of the following result:

Lemma 2.5 [Bloch and Esnault 2004, Lemma 3.3]. *Let M be a connection on K . The slopes of M are all ≤ 1 (respectively ≥ 1) if and only if there exists a good lattice pair $\mathcal{V}, \mathcal{W}$ such that $\mathcal{W} \subseteq (1/t)\mathcal{V}$ (respectively $\mathcal{W} \supseteq (1/t)\mathcal{V}$).*

(Note that the original proof by Bloch and Esnault needs the assumption that K is complete.)

3. Stationary phase principle

Let $K = k(t')$. For any scheme X over k and any $\mathbb{O}_X$ -modules $\mathcal{F}$, let $X_K = X \otimes_k K$ and $\mathcal{F}_K = \mathcal{F}|_{X_K}$. For any k -morphism $f : X \rightarrow Y$, let $f_K : X_K \rightarrow Y_K$ be the base change of f .

We keep the notation used in Section 1. In this section we prove Theorem 1.1.

For any $x \in T_K = T$, $(\mathcal{V}_K)_x, (\mathcal{W}_K)_x$ is a good lattice pair of the connection $(j_{K*}\mathcal{M}_K)_x$ on $K(t_x)$. Since t' is not algebraic over k , by Lemma 2.3, we may assume that

$$\mathcal{V}_\infty \otimes_{\mathbb{O}_{\mathbb{P}_K^1, \infty}} \mathbb{O}_{\mathbb{P}_K^1, \infty}, \left(\mathcal{W}_\infty + \frac{1}{z} \mathcal{V}_\infty \right) \otimes_{\mathbb{O}_{\mathbb{P}_K^1, \infty}} \mathbb{O}_{\mathbb{P}_K^1, \infty}$$

is a good lattice pair of the connection

$$\partial_z - \frac{t'}{z^2} : (j_{K*}\mathcal{M}_K)_\infty \rightarrow (j_{K*}\mathcal{M}_K)_\infty.$$

Lemma 3.1. *The inclusion of complexes $(1-1) \subset (*)$ induces a quasi-isomorphism*

$$(1-1)|_{\mathbb{P}_K^1} \simeq (*)|_{\mathbb{P}_K^1}.$$

Proof. We have

$$\begin{aligned} (1-1)|_{\mathbb{P}_K^1} &= (\mathcal{V}_K \xrightarrow{j_{K*}\nabla_K + t'dt} \Omega_{\mathbb{P}_K^1}^1(T_K) \otimes (\mathcal{W}_K + \mathcal{V}_K(\{\infty\}))), \\ (*)|_{\mathbb{P}_K^1} &= (j_{K*}\mathcal{M}_K \xrightarrow{j_{K*}\nabla_K + t'dt} \Omega_{\mathbb{P}_K^1}^1 \otimes j_{K*}\mathcal{M}_K). \end{aligned}$$

First we have $(1-1)|_{U_K} = (*)|_{U_K}$. For any $x \in S_K$, let's prove $(1-1)|_{\mathbb{P}_K^1} \subset (*)|_{\mathbb{P}_K^1}$ induces a quasi-isomorphism on the stalks at x . It suffices to show that

$$\left(\frac{1}{t_x^i} (\mathcal{V}_K)_x \Big/ \frac{1}{t_x^{i-1}} (\mathcal{V}_K)_x \right) \xrightarrow{\text{gr}_i(\partial_{t_x} + t')} \left(\frac{1}{t_x^{i+1}} (\mathcal{W}_K)_x \Big/ \frac{1}{t_x^i} (\mathcal{W}_K)_x \right)$$

is an isomorphism for any $i \geq 1$. As $(\mathcal{V}_K)_x \subset (\mathcal{W}_K)_x$, the map $\text{gr}_i(\partial_{t_x} + t')$ is equal to $\text{gr}_i(\partial_{t_x})$, which is an isomorphism by the definition of good lattices. The inclusion

$$((1-1)|_{\mathbb{P}_K^1})_\infty \rightarrow ((*|_{\mathbb{P}_K^1})_\infty$$

can be written as

$$\begin{aligned} \left(\mathcal{V}_\infty \otimes_{\mathbb{O}_{\mathbb{P}_k^1, \infty}} \mathbb{O}_{\mathbb{P}_K^1, \infty} \xrightarrow{\partial_z - \frac{t'}{z^2}} \frac{1}{z} \left(\mathcal{W}_\infty + \frac{1}{z} \mathcal{V}_\infty \right) \otimes_{\mathbb{O}_{\mathbb{P}_k^1, \infty}} \mathbb{O}_{\mathbb{P}_K^1, \infty} \right) \\ \subset ((j_{K*} \mathcal{M}_K)_\infty \xrightarrow{\partial_z - \frac{t'}{z^2}} (j_{K*} \mathcal{M}_K)_\infty). \end{aligned}$$

It is a quasi-isomorphism by the assumption on $\mathcal{V}_\infty$ and $\mathcal{W}_\infty$. $\square$

Lemma 3.2. $R^1 \bar{p}_{2*}(1-1)|_{\eta'} \cong R^1 \bar{p}_{2*}(\ast)|_{\eta'}.$

Proof. Consider the Cartesian diagram

$$(3-1) \quad \begin{array}{ccc} \mathbb{P}_K^1 & \longrightarrow & \eta' = \text{Spec } K \\ \downarrow & & \downarrow \\ \mathbb{P}_k^1 \times \mathbb{P}_k^1 & \xrightarrow{\bar{p}_2} & \mathbb{P}_k^1. \end{array}$$

By Lemma 3.1, we have

$$R^1 \bar{p}_{2*}(1-1)|_{\eta'} \cong H^1(\mathbb{P}_K^1, (1-1)|_{\mathbb{P}_K^1}) \cong H^1(\mathbb{P}_K^1, (\ast)|_{\mathbb{P}_K^1}) \cong R^1 \bar{p}_{2*}(\ast)|_{\eta'}. \quad \square$$

Corollary 3.3. $\mathcal{F}(i_* \mathcal{M})|_{\eta'_\infty} = R^1 \bar{p}_{2*}(1-1)|_{\eta'_\infty}.$

Denote by $\mathbb{P}_k^1[[z']]$ the formal completion of $\mathbb{P}_k^1 \times \mathbb{P}_k^1$ along its closed subscheme $\mathbb{P}_k^1 \times \{\infty\}$. For any coherent sheaf $\mathcal{H}$ on $\mathbb{P}_k^1$, let $\mathcal{H}[[z']] = \mathcal{H}|_{\mathbb{P}_k^1[[z']]}$.

Lemma 3.4 [Bloch and Esnault 2004, Corollary 2.2].

$$R^1 \bar{p}_{2*}(1-1) \otimes_{\mathbb{O}_{\mathbb{P}_k^1}} k[[z']] \cong H^1\left(\mathbb{P}^1[[z']], \mathcal{V}[[z']] \xrightarrow{z' \nabla + dt} \Omega_{\mathbb{P}^1}^1(T) \otimes \mathcal{W}[[z']]\right).$$

Lemma 3.5 [Bloch and Esnault 2004, Lemma 2.4 and Corollary 2.5]. *Let $\mathcal{H}$ be the complex*

$$\mathcal{V}[[z']] \xrightarrow{z' \nabla + dt} (\Omega_{\mathbb{P}_k^1}^1(T) \otimes (\mathcal{W} + \mathcal{V}(\{\infty\})))[[z']].$$

Then $\mathcal{H}^0$ equals (0) and $\mathcal{H}^1$ is supported on $T \subset \mathbb{P}_k^1 = \mathbb{P}_k^1[[z']]$. For any $x \in T$, let $\widehat{\mathcal{V}}_x = \mathcal{V}_x \otimes_{\mathbb{O}_{\mathbb{P}_k^1, x}} k[[t_x]]$ and $\widehat{\mathcal{W}}_x = \mathcal{W}_x \otimes_{\mathbb{O}_{\mathbb{P}_k^1, x}} k[[t_x]]$. We have

$$\mathcal{H}_x^1 = \text{coker}\left(\widehat{\mathcal{V}}_x[[z']] \xrightarrow{z' \nabla + dt} \Omega_{\mathbb{P}_k^1}^1(T) \otimes (\widehat{\mathcal{W}}_x + \widehat{\mathcal{V}}(\{\infty\})_x)[[z']]\right). \quad \square$$

Corollary 3.6.

$$\begin{aligned} H^1(\mathbb{P}_k^1[[z']], \mathcal{H}) &= \bigoplus_{x \in S} \text{coker}\left(\widehat{\mathcal{V}}_x[[z']] \xrightarrow{z' \partial_{t_x} + 1} \frac{1}{t_x} \widehat{\mathcal{W}}_x[[z']]\right) \\ &\quad \oplus \text{coker}\left(\widehat{\mathcal{V}}_\infty[[z']] \xrightarrow{z' \partial_z - \frac{1}{z^2}} \frac{1}{z} \left(\widehat{\mathcal{W}}_\infty + \frac{1}{z} \widehat{\mathcal{V}}_\infty \right)[[z']]\right). \end{aligned}$$

Combining Corollary 3.3, Lemma 3.4 and Corollary 3.6, we have

$$\begin{aligned} \mathcal{F}(i_*\mathcal{M})|_{\eta'_\infty} &= R^1\bar{p}_{2*}(1-1) \otimes_{\mathbb{O}_{\mathbb{P}^1_k}} k[[z']] \otimes_{k[[z']] } k((z')) \\ &= \bigoplus_{x \in S} \operatorname{coker}(\widehat{\mathcal{V}}_x((z')) \xrightarrow{z'\partial_{t_x}+1} \frac{1}{t_x}\widehat{\mathcal{W}}_x((z'))) \\ &\quad \oplus \operatorname{coker}(\widehat{\mathcal{V}}_\infty lpz') \xrightarrow{z'\partial_z-\frac{1}{z^2}} \frac{1}{z}(\widehat{\mathcal{W}}_\infty + \frac{1}{z}\widehat{\mathcal{V}}_\infty)((z'))). \end{aligned}$$

The left side of this equality is independent of the choice $\mathcal{V}$ and $\mathcal{W}$. For any $i \in \mathbb{N}$, $\mathcal{V}(iT)$ and $\mathcal{W}(iT)$ still satisfy the condition of Lemma 3.1. Then the above equality holds if we replace $\mathcal{V}$ and $\mathcal{W}$ by $\mathcal{V}(iT)$ and $\mathcal{W}(iT)$, respectively. Taking the direct limit on i , we have

$$\begin{aligned} \mathcal{F}(i_*\mathcal{M})|_{\eta'_\infty} &= \bigoplus_{x \in S} \operatorname{coker}((\mathcal{M}|_{\eta_x})((z')) \xrightarrow{z'\partial_{t_x}+1} (\mathcal{M}|_{\eta_x})((z'))) \\ &\quad \oplus \operatorname{coker}((\mathcal{M}|_{\eta_\infty})((z')) \xrightarrow{z'\partial_z-\frac{1}{z^2}} (\mathcal{M}|_{\eta_\infty})((z'))). \end{aligned}$$

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References

- [Bloch and Esnault 2004] S. Bloch and H. Esnault, “Local Fourier transforms and rigidity for $\mathcal{D}$ -modules”, *Asian J. Math.* **8**:4 (2004), 587–605. MR 2006b:14028 Zbl 1082.14506
- [Deligne 1970] P. Deligne, *Équations différentielles à points singuliers réguliers*, Lecture Notes in Mathematics **163**, Springer, Berlin, 1970. MR 54 #5232 Zbl 0244.14004
- [Katz 1988] N. M. Katz, “Travaux de Laumon”, pp. 105–132, exposé 691 in *Séminaire Bourbaki, 1987/88*, Astérisque **161-162**, Soc. Mat. de France, Paris, 1988. In English. MR 90h:14028 Zbl 0698.14014
- [Laumon 1987] G. Laumon, “Transformation de Fourier, constantes d’équations fonctionnelles et conjecture de Weil”, *Inst. Hautes Études Sci. Publ. Math.* **65** (1987), 131–210. MR 88g:14019

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